# Arista Access Node IND-ACC-16 Hardware/Environmental Fault Resolution SOP

**Node:** IND-ACC-16 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_8780

## Description

This Standard Operating Procedure (SOP) is designed to address hardware/environmental faults on Arista Access Node IND-ACC-16, specifically ASIC memory parity errors, line card crashes, and power supply redundancy loss, as indicated by alarm code ALARM_8780. The Arista 7250X series switch is a high-performance access node that utilizes a multi-chassis architecture to provide redundant power supply and line card capabilities. However, due to the intricate nature of the system, hardware and environmental faults can occur, leading to loss of redundancy and impacting network availability. In this SOP, we will walk through the diagnostic and resolution steps to address the root cause of the issue, ensuring minimal downtime and optimal network performance.

## Pre-requisite Checks

- Check: Verify the Indiranagar Exchange site is accessible and secure for remote hands assistance, Status: Complete

- Check: Confirm the redundant link status on the Arista access node, Status: Complete

- Check: Review the system logs for any recent hardware or environmental issues, Status: Complete

## Potential Root Causes

| Cause | Probability |
|---|---|
| ASIC memory parity error | High |
| Line card crash | Medium |
| Power supply redundancy loss | Low |

## Diagnostic Commands & Expected Outputs

```
Command: show system internal hardware-module, Expected Output: Hardware module information, including presence and status, Description: This command is used to verify the presence and status of the hardware modules, including the line cards and power supplies, on the Arista access node.

Command: show system internal software-module, Expected Output: Software module information, including version and status, Description: This command is used to verify the software version and status on the Arista access node, including the operating system and firmware versions.

Command: show system internal alarm, Expected Output: Alarm information, including reason and status, Description: This command is
```

```
used to verify the alarm information on the Arista access node,
including the reason for the alarm and its current status.
```

## Resolution Steps

Step 1: Step: Replace the faulty ASIC, Command: replace system hardware-module , Description: Replace the faulty ASIC with a new one, ensuring that the replacement is done in a timely manner to minimize downtime.

Step 2: Step: Reboot the line card, Command: reboot system hardware-module , Description: Reboot the faulty line card to resolve the issue, ensuring that the reboot is done in a controlled manner to minimize impact on the network.

Step 3: Step: Replace the power supply, Command: replace system power-supply , Description: Replace the faulty power supply with a new one, ensuring that the replacement is done in a timely manner to minimize downtime.

## Post-Check Verification

```
Command: show system internal hardware-module, Expected Output:
Hardware module information, including presence and status,
Description: Verify that the hardware module has been replaced or
fixed, and that the issue has been resolved.

Command: show system internal software-module, Expected Output:
Software module information, including version and status,
Description: Verify that the software version and status on the Arista
access node have been updated, and that the issue has been resolved.

Command: show system internal alarm, Expected Output: Alarm
information, including reason and status, Description: Verify that the
alarm information on the Arista access node has been cleared, and that
the issue has been resolved.
```

## Roll-back Procedures

- Step: Restore the original configuration, Command: restore system configuration , Description: Restore the original configuration to the Arista access node to revert any changes made during the resolution process.

- Step: Reboot the system, Command: reboot system, Description: Reboot the system to ensure that any changes made during the resolution process are fully applied and that the system is stable.